

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (ME) Examination****Nov-Dec 2018****ME244 Manufacturing Processes - I****Date: 28.11.2018, Wednesday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.****[07]**

- i) Which of the following pattern is used to produce a number of castings in sand casting?
 - a) Loose piece pattern
 - b) Split pattern
 - c) Gated pattern
 - d) Match plate pattern
- ii) Which of the following is negative type allowance?
 - a) Shrinkage
 - b) Shake
 - c) Draft
 - d) Machining
- iii) The function of cores in casting is to
 - a) form extended parts
 - b) form internal cavities
 - c) used for directional solidification
 - d) none of the above
- iv) Fly wheels, track wheels and gears can be produced mainly by
 - a) Investment casting
 - b) Continuous casting
 - c) Die casting
 - d) Centrifugal casting
- v) What is a function of a riser in a sand casting?
 - a) Provide good mobility to the molten metal
 - b) Prevent cavities due to shrinkage
 - c) To develop holes or hollow cavities in the casting
 - d) To filter the molten metal
- vi) Which one is the most widely used resin in shell moulding?
 - a) Phenol formaldehyde
 - b) Ethanol
 - c) Colloidal Silica
 - d) Clay
- vii) Which of the following is not true for casting inspection?
 - a) Depth of the defect can be obtained by Ultrasonic testing.
 - b) X-rays are used to detect defect in Radiography test.
 - c) Dye penetrant test is useful to determine internal defects.
 - d) Magnetic particle testing can be used for steel castings.

- Q – 2 (a)** Classify basic manufacturing processes and explain it in brief. **[04]**
- (b)** List various melting furnaces. Explain cupola furnace. **[05]**
- (c)** Represent mould making procedure of investment casting process with sketch only. **[05]**
State any three key advantages of investment casting process.

- Q – 3 (a)** Define a pattern. Discuss how a pattern differs from a casting. Explain any two types of pattern with sketch. **[07]**
- (b)** Enlist the functions of gating system. Draw necessary sketch representing various components of gating system. **[04]**
- (c)** State one major cause and one prime remedy of following sand casting defects. **[03]**
1) Flash 2) Gas porosity 3) Sand inclusion

OR

- Q – 3 (a)** Which are different pattern allowances? Explain any two of them with sketch. **[04]**
- (b)** Explain metal die casting process with neat sketch. Discuss advantages of die casting process over sand casting. **[06]**
- (c)** Which are the different properties of moulding sand needed to be controlled for its effective performance? Discuss any one sand control test used to assess moulding sand mixture. **[04]**

SECTION – II

Q - 4 Do As Directed.

- i)** Define: Plastic welding and fusion welding process. **[02]**
- ii)** Explain the term “Flux” used in welding process. **[01]**
- iv)** “Higher amount of heat is produced in second stage combustion of fuel gases in oxy-acetylene welding”. Justify the statement. **[02]**
- v)** Which of the following welding techniques uses a non-consumable electrode? Chose correct answer. **[01]**
- a) MIG welding b) TIG welding c) Submerged arc welding d) Thermit welding
- vi)** Explain Heat Affected Zone (HAZ) in case of welding process. **[01]**

- Q – 5 (a)** Classify welding processes in detail. Describe common applications of welding. **[06]**
- (b)** List various fuel gases used in oxy-fuel gas welding processes. Describe oxy-acetylene welding equipment with sketch. **[06]**
- (c)** What is the function of flux in Submerged Arc Welding (SAW) process? **[02]**

OR

- Q – 5 (a)** Describe Shielded Metal Arc Welding (SMAW) process with sketch. Discuss advantages, disadvantages of SMAW. **[06]**
- (b)** What does electrode code PXXXXXXS indicates? Discuss as per Indian standard (IS). **[06]**
- (c)** How gas cutting torch differs from a gas welding torch? **[02]**

- Q – 6 (a)** Discuss Metal Inert Gas (MIG) welding technique with sketch. State applications of It. **[06]**
- (b)** Differentiate between electron beam welding and laser beam welding process. **[06]**
- (c)** What are the gases used in Gas Metal Arc Welding (GMAW) process? **[02]**

OR

- Q – 6 (a)** Describe ultrasonic welding process with sketch. State its applications. **[06]**
- (b)** Explain basic principle of resistance welding. Discuss seam welding process with suitable diagram. **[06]**
- (c)** Explain explosive welding process in brief. **[02]**
